

Post-Quantum Security

Why it matters, what the risks are, and what action is needed?

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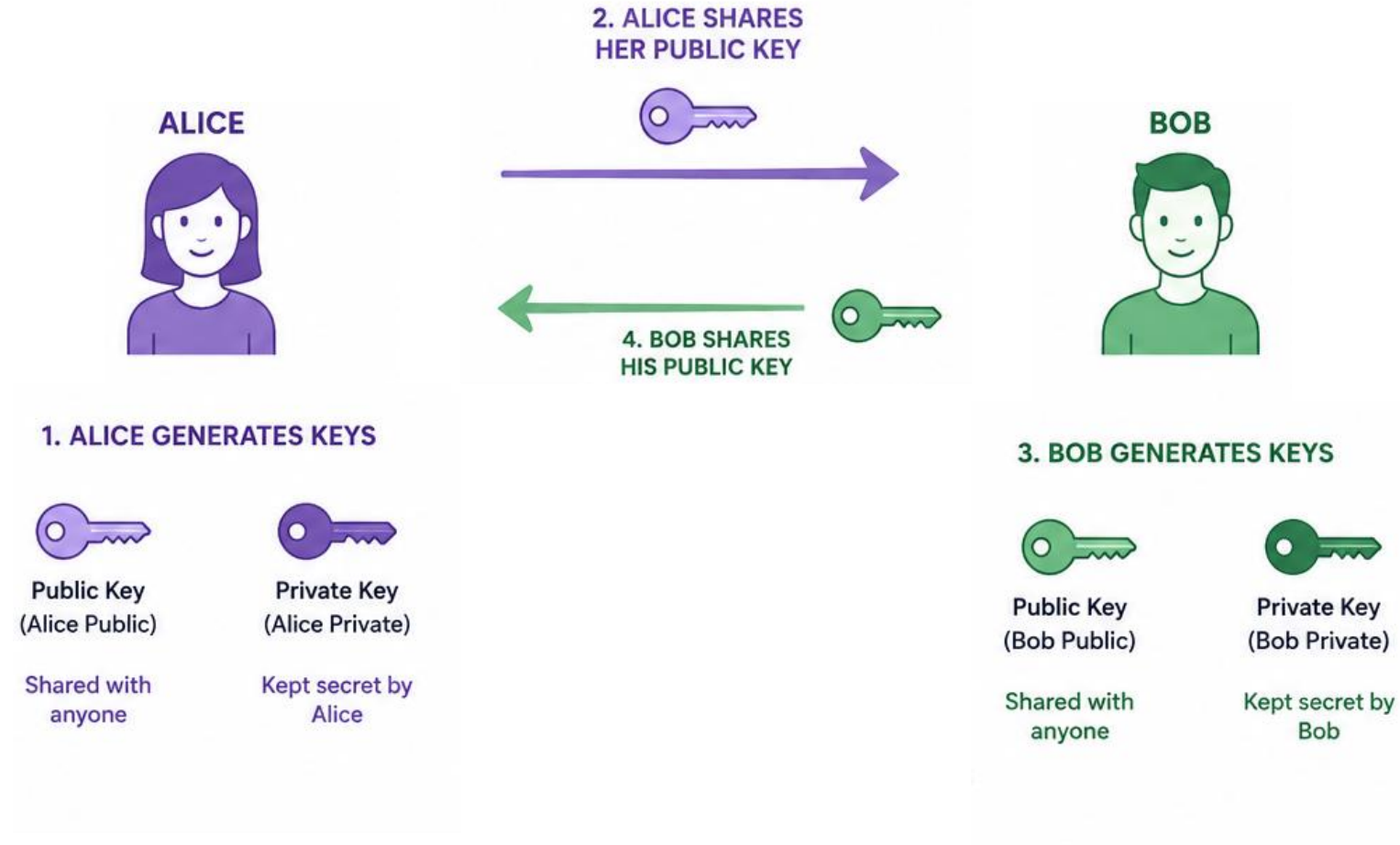


Loughborough
University



UK
Parliament

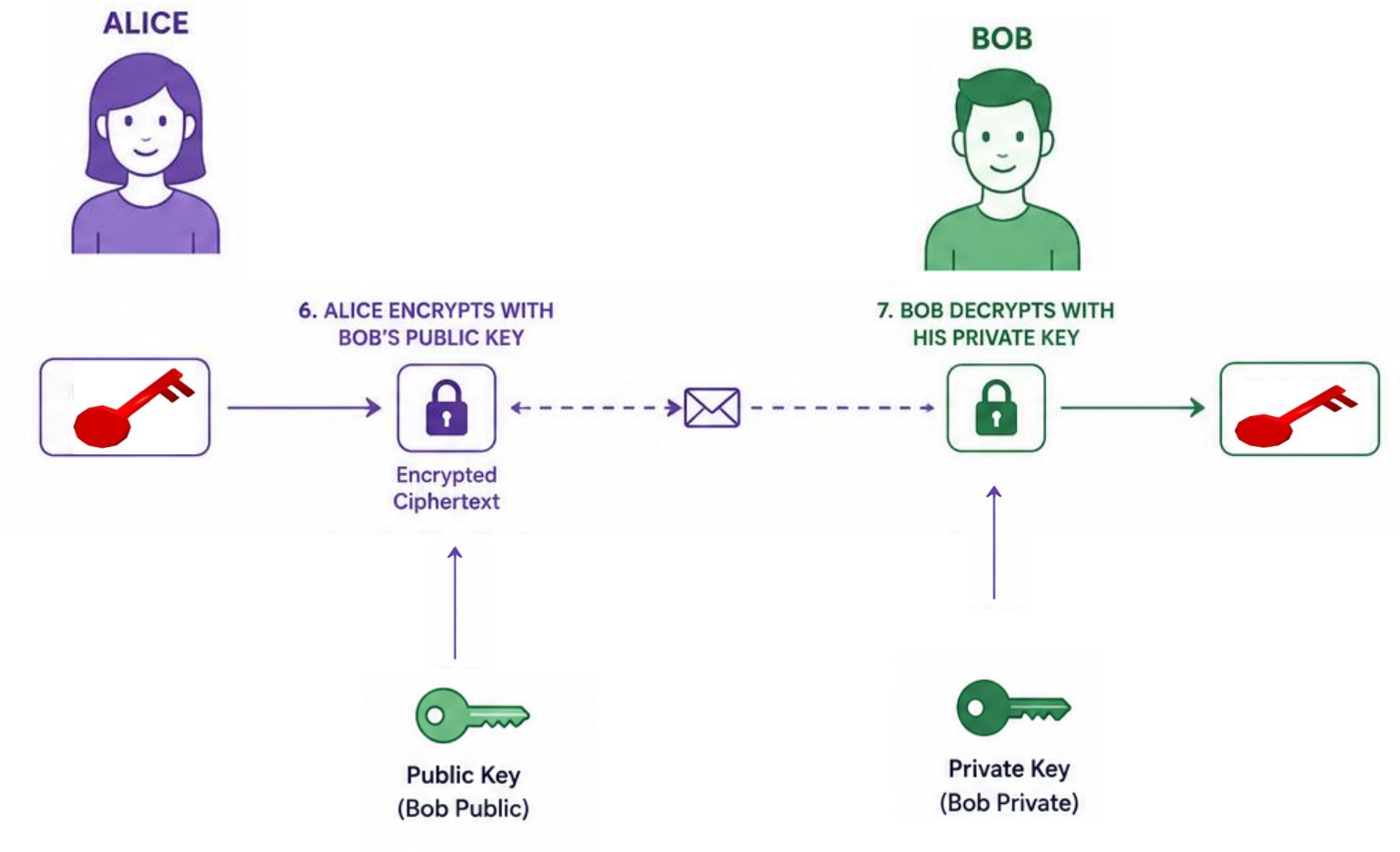
What is public-key cryptography?



What is public-key cryptography?



One key encrypts & other key decrypts



What happens now when we send secrets over the internet?

Step 1



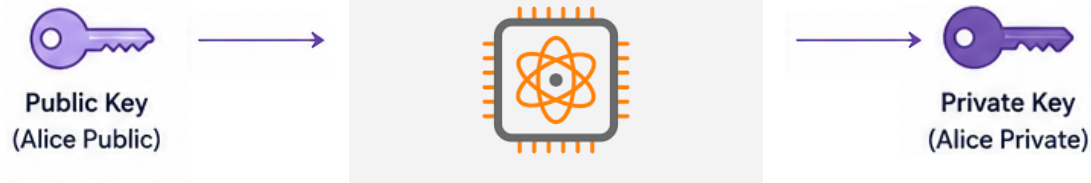
Agree on
secret
key for
future
sessions

Step 2



What is the problem?

**Quantum
Computers
Can break this**



Dr. Peter Shor, American Mathematician and MIT Professor who devised Shor Algorithm in 1994

Why can quantum computers do this?

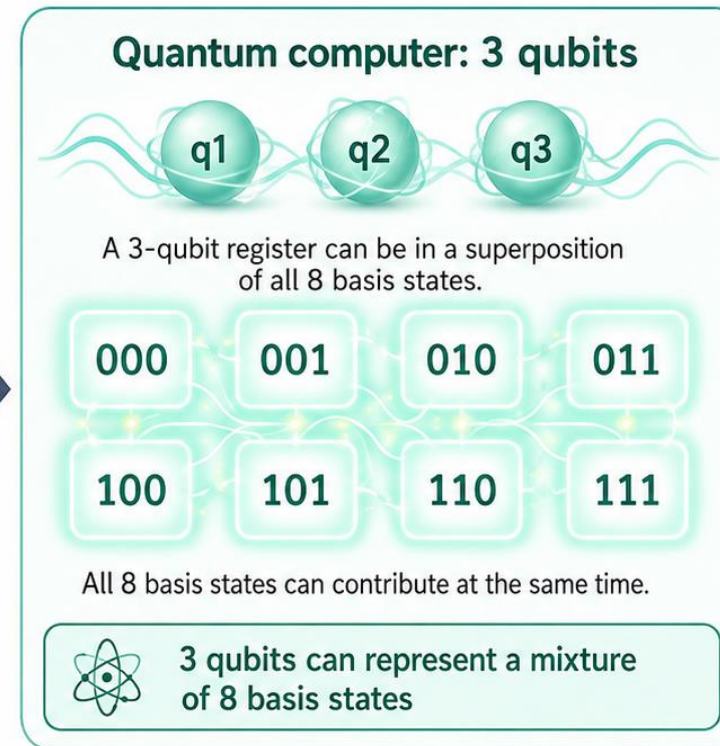
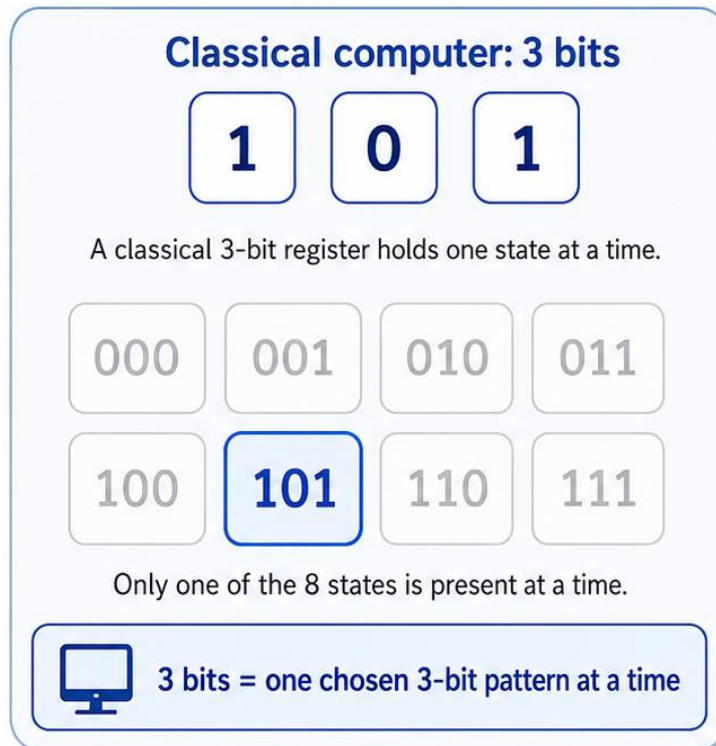
- Classical Computers use trial and error to break the secret
 - Check the possibilities one-by-one
- Quantum computers exploit quantum physics to compare multiple combinations at a time
 - This will exponentially increase the speed of searching the secret



Public Key
(Alice Public)

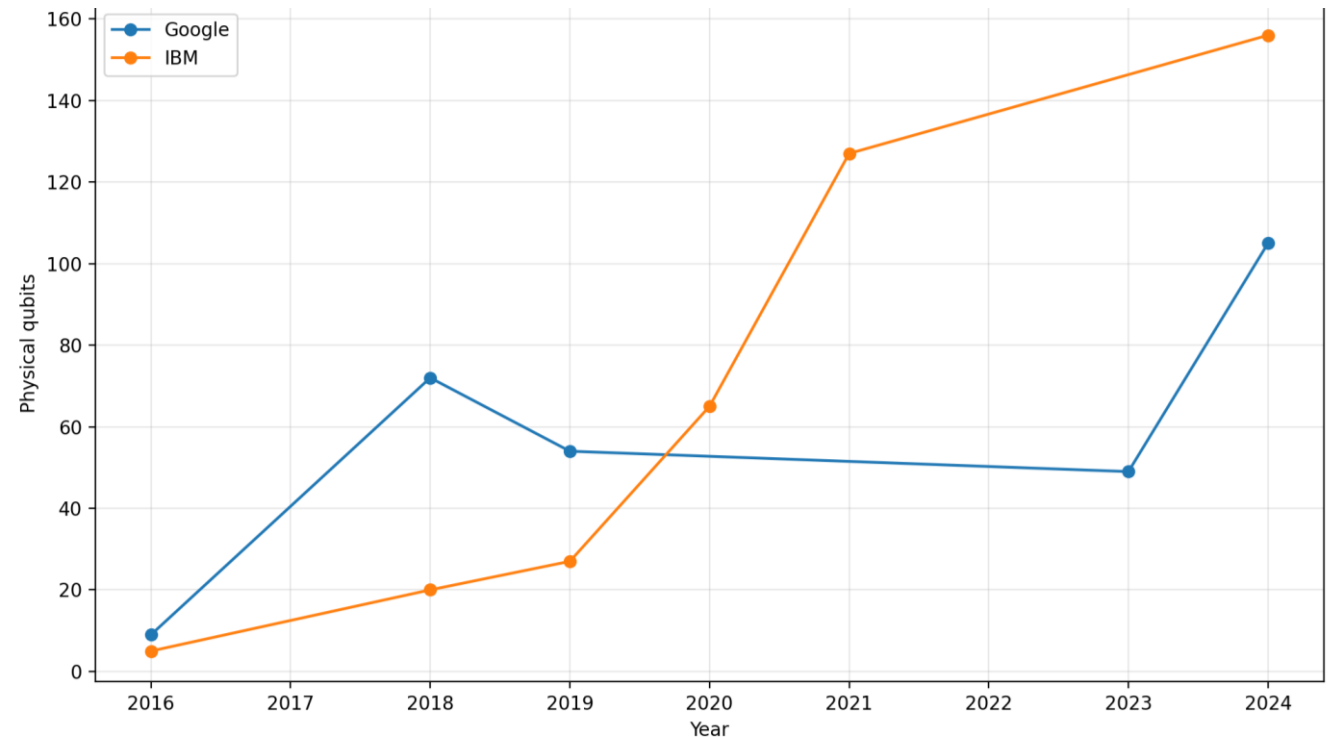
= P x Q

Digital Bits (classical computers) vs Qubits (quantum computers)



Current capabilities (based on **physical qubits**)

- IBM - 120 and 156 qubits QC(exp. 1080 by 2028)
- Google - 105 qubits QC
- D-Wave Advantage2 - 1044 qubits QC
- ...



What is needed to break RSA/ECC?

- **RSA2048** – 6000 logical qubits (est. 20 million physical qubits as of 2021; less than 1 million physical qubits as of 2025)
- **ECC256** – around 1500 logical qubits (less than 0.5 million physical qubits; there are some research suggest 10,000 physical qubits could be enough for this as well)

[quant-ph] 30 Mar 2026

Shor's algorithm is possible with as few as 10,000 reconfigurable atomic qubits

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Quantum computers have the potential to perform computational tasks beyond the reach of classical machines. A prominent example is Shor's algorithm for integer factorization and discrete logarithms, which is of both fundamental importance and practical relevance to cryptography. However, due to the high overhead of quantum error correction, optimized resource estimates for cryptographically relevant instances of Shor's algorithm require millions of physical qubits. Here, by leveraging advances in high-rate quantum error-correcting codes, efficient logical instruction sets, and circuit design, we show that Shor's algorithm can be executed at cryptographically relevant scales with as few as 10,000 reconfigurable atomic qubits. Increasing the number of physical qubits improves time efficiency by enabling greater parallelism; under plausible assumptions, the runtime for discrete logarithms on the P-256 elliptic curve could be just a few days for a system with 26,000 physical qubits, while the runtime for factoring RSA-2048 integers is one to two orders of magnitude longer. Recent neutral-atom experiments have demonstrated universal fault-tolerant operations below the error-correction threshold, computation on arrays of hundreds of qubits, and trapping arrays with more than 6,000 highly coherent qubits. Although substantial engineering challenges remain, our theoretical analysis indicates that an appropriately designed neutral-atom architecture could support quantum computation at cryptographically relevant scales. More broadly, these results highlight the capability of neutral atoms for fault-tolerant quantum computing with wide-ranging scientific and technological applications.

Quantum computers are believed to be capable of solving certain problems that are intractable for classical machines [1, 2]. A prominent example is Shor's algorithm

algorithm—typically require on the order of millions of physical qubits [18]. This scale is driven by the large overhead of quantum error correction, which often re-

What is the problem?

- Migration to post-quantum security will take years
- The main risk is “**harvest now, decrypt later**”
- Energy
 - transaction -10 years
 - Nuclear – 70 years
- Finance – 120 years
- Healthcare - 120 years
- Transport – 80 years
- National security – forever?

What the alternatives are?



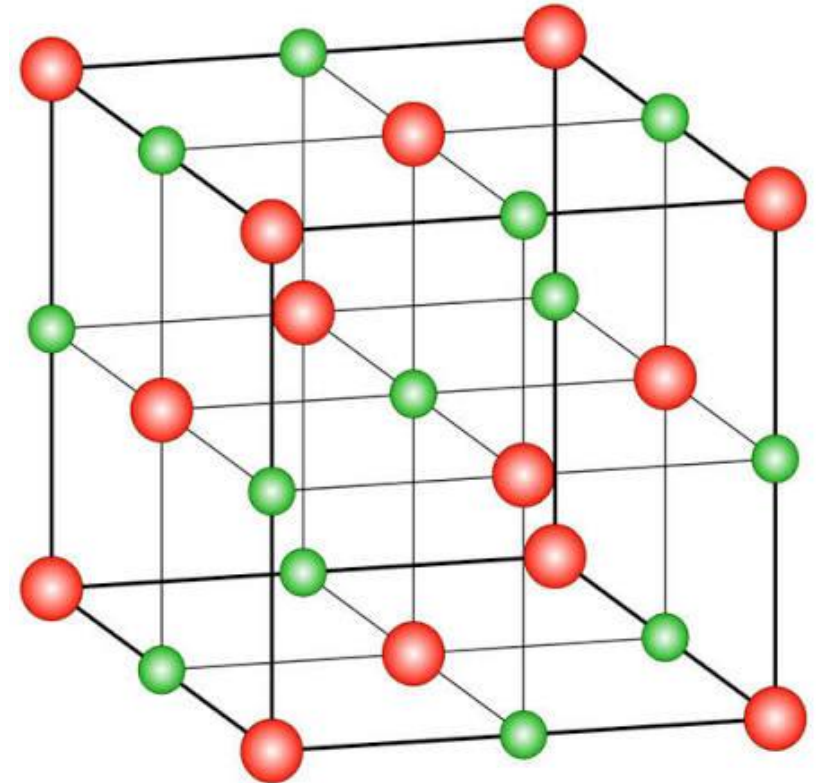
**Post-quantum
cryptography (PQC)**



Quantum key distribution

Post-quantum cryptography (PQC)

- Introduced in 1996
- NIST recommended 5 candidates
- 3 out 5 candidates are based on lattices – aka Lattice-based Crypto
- It relies on geometry – a different types mathematics



Who get the capability first?

- **States** — most likely to have early strategic capability due to budgets, secrecy, and national security incentives
- **Large technology companies (IBM, Google, etc)** — key role in building components, platforms, and talent
- **Criminal actors** — unlikely to lead development

NCSC timeline (March 2025) for the UK

Compared to other countries

	Initial Plan	Migration	Completion
The UK	2028	2031	2035
EU	2026	2028	2030
Australia	2026	2028	2030
Canada	2028	2031	2035
United States	Now	ASAP	ASAP

Who is leading the activities in the UK

- NCSC
 - **PQC Pilot** - NCSC's Assured Cyber Security Consultancy Post-Quantum Cryptography pilot (up until 2027)
- DSIT
- Crown Commercial Services
- The following 8 companies are part of the NCSC PQC Pilot:
 - Arqit; CGI IT UK Limited; Avella Security Limited; Cambridge Consultants Ltd.; Capgemini; Frazer-Nash Consultancy; IBM; PQShield

What should government and policy makers do?

- We need to address the uncertainty on Q-day
- Reports targeting C-level
- Work with academia and industry to develop training programmes
- Publish intensive training courses
- Clear cost-benefit analysis
- Specific regulation action to mandate PQC

Closing Remarks

- Quantum threat to current public-key cryptography is credible, even if timing is uncertain
- The hardest part is not inventing algorithms — it is migrating real systems at scale
- Preparation must begin before the threat is fully visible